EXPERIMENT 14

Exploring Pro-Code Extension of Zoho Creator Using Custom Functions

Aim

Extend a low-code application with reusable Deluge logic and demonstrate input validation, return values and workflow reuse.

Record status

Custom-function route documented using existing saved code and earlier live tests. No Catalyst service or new function deployment is claimed.

Concept

The exercise allows a custom-function route. A function has a name, argument types, return type and optional namespace. It centralizes logic that multiple workflow types can call. The Insurance Claim application already contains a settlement-letter function with direct execution and Blueprint invocation evidence.

Procedure

1. In Workflow, Functions, inspect the settlement function and its typed claimID argument. Retain the existing tested implementation.

2. Explain the function contract: accept a claim identifier and return a clear success, skip or error message. Fetch only the requested record.

3. Validate record existence, claimant name and positive payable amount before producing a letter. Keep sensitive delivery actions behind status and authorization checks.

4. Call the function directly with a valid fictional claim and invalid ID. Then inspect its call from the Blueprint settlement action.

5. Compare the stored PDF and audit entries with the requested claim. Keep delivery acceptance distinct from inbox receipt and keep the payout simulation explicitly labelled.

Application design and observations

Function element	Application example
Input	Integer claimID
Return	String result for a caller or execution log
Record access	Claim_Request_Form[ID == claimID]
Reuse	Direct execution and Blueprint settlement
Current delivery helper	GenerateSettlementLetterWithDelivery
Optional future extension	Catalyst or external service behind an authenticated interface

Deluge example

```
// Representative validation pattern from the earlier function.
claim = Claim_Request_Form[ID == claimID];
if(claim.count() == 0)
{
    return "ERROR: Claim record not found.";
}
amount = ifnull(claim.Net_Payable_Amount, 0);
if(amount <= 0)
{
    return "ERROR: Positive payable amount required.";
}
// A caller can invoke the existing function as follows:
// result = thisapp.GenerateSettlementLetter(claimID);
```

The code above is an explanatory excerpt, not a replacement for the current helper. The current workflow generates and stores a PDF and checks settlement state before delivery. The earlier standalone function source stored locally predates that helper and should be treated as an archived teaching example.

Validation and discussion

Test	Earlier observed result
Valid fictional record	PDF generated and stored
Invalid record ID	Controlled error returned
Blueprint settlement call	Final PDF stored and email request queued
Automatic sequential duplicate	Amount/recipient audit guard configured
Explicit resend	Saved but its email execution remains uncompleted

Result

The existing application demonstrates pro-code extension through reusable Deluge functions. Direct function execution and the Blueprint call have earlier live evidence. Catalyst hosting and external service deployment are optional future work, outside this custom-function route.

Viva focus

Explain typed arguments, return values, function reuse and input validation.

Reading: Create a Deluge function (official Zoho documentation, reviewed 19 October 2026).

Reading: Call a function in Creator (official Zoho documentation, reviewed 19 October 2026).